

NetComms 2 Module 4 Reviewer: Understanding and Using APIs

DevNet Associate, Module 4. Built from DEVASC_Module_4.pdf. Choice and matching only.

Instructions: choose the best answer for every item. Matching items use a dropdown on the real exam, so pick one letter for each term. The answer key is at the end.

Concept Questions (Slide Order)

1. What does an API allow?
 - A. A database to compress data
 - B. Hardware to boot faster
 - C. One piece of software to talk to another
 - D. A user to install an OS

2. Which is a use case of APIs given in the module?
 - A. Hardware manufacturing
 - B. Battery testing
 - C. Automation tasks
 - D. Manual data entry

3. Synchronous APIs respond to a request how?
 - A. With a response with no data
 - B. Only after polling
 - C. Directly, providing data immediately
 - D. Only after a webhook

4. What does an asynchronous API return to signify the request was received?
 - A. A webhook URL
 - B. The full data
 - C. A response with no data
 - D. An error code

5. What is the benefit of an asynchronous API design?
- A. The application can continue without being blocked while the server processes the request
 - B. No client is needed
 - C. Data is returned immediately
 - D. The application must wait for the response
6. Which are the three most popular API architectural styles?
- A. REST, GraphQL, and gRPC
 - B. HTTP, FTP, and SMTP
 - C. SOAP, XML, and JSON
 - D. RPC, SOAP, and REST
7. Which API style is a request-response model that lets an application make a procedure call to another application?
- A. Remote Procedure Call (RPC)
 - B. GraphQL
 - C. REST
 - D. Webhook
8. Which are the elements of a SOAP message?
- A. Prologue, Header, Body, and Log
 - B. Envelope, Route, Body, and Tail
 - C. Envelope, Header, Body, and Fault
 - D. Head, Body, Footer, and Error
9. Which SOAP property means it can be used over any protocol, such as HTTP, SMTP, TCP, UDP, or JMS?
- A. Stateless
 - B. Independent
 - C. Extensible
 - D. Neutral

10. Who authored the REST architectural style?

- A. Erich Gamma
- B. Roy Thomas Fielding
- C. Alistair Cockburn
- D. Winston Royce

11. How many constraints did the author establish for REST?

- A. Four
- B. Six
- C. Three
- D. Twelve

12. Which REST constraint says the server cannot contain session states?

- A. Code-on-demand
- B. Stateless
- C. Cache
- D. Layered system

13. Which REST constraint states responses must say whether they are cacheable?

- A. Client-server
- B. Stateless
- C. Cache
- D. Uniform interface

14. Which is one of the four principles of the REST uniform interface?

- A. Layered proxies
- B. Executable code in responses
- C. Session state on the server
- D. Hypermedia as the engine of application state

15. Which REST constraint is optional because of the security risks of third-party code?

- A. Code-on-demand
- B. Stateless
- C. Client-server
- D. Cache

16. What are the four major components of a REST API request?

- A. Scheme, host, port, and path
- B. URI, HTTP method, header, and body
- C. Status, header, body, and cookie
- D. URL, token, key, and query

17. Which URI component specifies whether http or https is used?

- A. Scheme
- B. Authority
- C. Query
- D. Path

18. Which URI component provides additional details for scope, filtering, or to clarify a request?

- A. Scheme
- B. Query
- C. Authority
- D. Path

19. Which URI component is the location of the resource on the server?

- A. Path
- B. Scheme
- C. Query
- D. Authority

20. Which HTTP method is suggested for the Create action?

- A. GET
- B. PUT
- C. POST
- D. DELETE

21. Which HTTP method is suggested for a partial update of an existing resource?

- A. POST
- B. PATCH
- C. PUT
- D. GET

22. Which REST API requests typically include a body?

- A. GET only
- B. POST, PUT, and PATCH
- C. GET, DELETE, and HEAD
- D. DELETE only

23. If data is provided in the body, which header key specifies its type?

- A. Cache-Control
- B. Content-Type
- C. Accept-Encoding
- D. Authorization

24. Which are the three major components of a REST API response?

- A. URI, method, and body
- B. Scheme, host, and path
- C. HTTP status, header, and body
- D. Cookie, token, and key

25. What does the 1xx status category mean?

- A. Success
- B. Redirection
- C. Informational
- D. Client error

26. What does the 3xx status category mean?

- A. Server error
- B. Informational
- C. Success
- D. Redirection

27. Which status code means the request was fulfilled and a resource was created?

- A. 204 No Content
- B. 200 OK
- C. 201 Created
- D. 202 Accepted

28. Which status code means the request was understood but rejected by the server?

- A. 401 Unauthorized
- B. 403 Forbidden
- C. 400 Bad Request
- D. 404 Not Found

29. Which status code means the server currently cannot handle the request?

- A. 500 Internal Server Error
- B. 502 Bad Gateway
- C. 503 Service Unavailable
- D. 404 Not Found

30. Which response header is used by the server to send cookies?

- A. Authorization
- B. Set-Cookie
- C. Content-Type
- D. Cache-Control

31. Which response header specifies directives that must be obeyed by all caching mechanisms?

- A. Set-Cookie
- B. Content-Type
- C. Accept-Encoding
- D. Cache-Control

32. Response pagination lets data be broken into chunks. How do most APIs specify the page to return?

- A. A cookie
- B. The Authorization header
- C. A query parameter
- D. The request body

33. Which header does a request use to ask for compressed data?

- A. Content-Type
- B. Cache-Control
- C. Accept-Encoding
- D. Set-Cookie

34. What does authentication prove?

- A. What the user may access
- B. The user's identity
- C. How many requests are allowed
- D. The data format

35. What does authorization define?

- A. The user's access to perform the action on a resource
- B. The user's identity
- C. The URI scheme
- D. The compression format

36. Which authentication mechanism uses a token generated by an authentication server such as an Identity Service?

- A. Basic authentication
- B. Bearer authentication
- C. API key
- D. Cookie

37. What are the two types of API keys?

- A. Client and server
- B. Basic and bearer
- C. Public and private
- D. Read and write

38. Which statement about OAuth 2.0 is correct?

- A. It is backwards compatible with 1.0
- B. It sends credentials to the third-party app
- C. It only handles authentication
- D. It is not backwards compatible with OAuth 1.0

39. What is an API rate limit?

- A. The size of a response
- B. A way for a web service to control how many requests a user or application can make per unit of time
- C. The number of servers
- D. The speed of the network

40. Which is a purpose of rate limiting?

- A. Increase the response size
- B. Cache responses
- C. Protect against a Denial-of-Service (DoS) attack
- D. Remove authentication

41. Which rate limit algorithm processes queued requests at a fixed rate and rejects when the queue is full?

- A. Leaky bucket
- B. Token bucket
- C. Fixed window counter
- D. Sliding window counter

42. Which rate limit algorithm gives each user a defined number of tokens to use in a time increment?

- A. Sliding window counter
- B. Fixed window counter
- C. Token bucket
- D. Leaky bucket

43. Which rate limit algorithm assigns a counter to a fixed window of time?

- A. Token bucket
- B. Fixed window counter
- C. Sliding window counter
- D. Leaky bucket

44. Which rate limit algorithm counts requests from the beginning of the window to the current time when a new request is made?
- A. Sliding window counter
 - B. Leaky bucket
 - C. Token bucket
 - D. Fixed window counter
45. Which rate limit header gives the number of requests the requester can still make in the current window?
- A. X-Rate Limit-Limit
 - B. X-Rate Limit-Reset
 - C. X-Rate Limit-Remaining
 - D. Content-Length
46. Which HTTP status codes are most commonly used when the rate limit is exceeded?
- A. 301 or 302
 - B. 500 or 503
 - C. 200 OK or 201 Created
 - D. 429 Too Many Requests or 403 Forbidden
47. Why are webhooks also called reverse APIs?
- A. They send data backwards
 - B. Applications subscribe to the webhook server by registering with the provider
 - C. They only use GET
 - D. They need no server
48. Which requirement applies to an application consuming a webhook?
- A. It must use only XML
 - B. It must register a URI on the webhook provider
 - C. It must poll the provider every minute
 - D. It must run only once a day

49. Which Python exception in the module indicates a URI missing its scheme?

- A. TimeoutError
- B. SSLError
- C. MissingSchema
- D. ConnectionRefused

50. In Python requests, which parameter turns off certificate verification in a lab?

- A. cert=None
- B. verify=False
- C. ssl=False
- D. timeout=0

51. Which status code means the client must first authenticate with a proxy?

- A. 403 Forbidden
- B. 409 Conflict
- C. 407 Proxy Authentication Required
- D. 401 Unauthorized

52. Which status code means the request conflicts with the current state of the resource, for example an edit conflict?

- A. 404
- B. 400
- C. 409
- D. 415

53. Which status code is returned when the client sends a body format the server does not support, such as XML to a JSON-only server?

- A. 406 Not Acceptable
- B. 415 Unsupported Media Type
- C. 400 Bad Request
- D. 500 Internal Server Error

54. Which status code means the server, acting as a gateway or proxy, did not receive a timely response from an upstream server?

- A. 503 Service Unavailable
- B. 501 Not Implemented
- C. 504 Gateway Timeout
- D. 502 Bad Gateway

55. Which status code means the server does not support the functionality required to fulfill the request?

- A. 500 Internal Server Error
- B. 504 Gateway Timeout
- C. 502 Bad Gateway
- D. 501 Not Implemented

56. What are the three steps to troubleshoot errors listed in the module?

- A. Check the return code, check the response body, use a status code reference
- B. Add tokens, add keys, add cookies
- C. Change the method, change the URI, change the header
- D. Restart the server, reinstall Python, clear the cache

57. Can server-side issues be resolved from the API client side?

- A. Yes, by turning off verify
- B. Yes, by changing the URI
- C. Yes, by changing the method
- D. No, contact the administrator of the API server

58. Which codes are client-side errors and which are server-side errors?

- A. 4xx client-side, 5xx server-side
- B. 1xx client-side, 3xx server-side
- C. 4xx server-side, 5xx client-side
- D. 2xx client-side, 5xx server-side

Scenario Questions

59. Your script needs to fetch the current temperature from a public weather REST API over HTTPS. Which HTTP method should you use for this request?
- A. PUT
 - B. DELETE
 - C. POST
 - D. GET
60. Your application uploads a video for processing, which can take several minutes. You decide to use an API that immediately returns a receipt and processes the video in the background. Which API design style is most appropriate?
- A. Synchronous API
 - B. SOAP API
 - C. Asynchronous API
 - D. RPC API
61. When designing a REST API, you want to ensure it follows the stateless constraint. Which of the following best describes a stateless interaction?
- A. The server stores session data for each client between requests
 - B. Each request contains all information needed to understand and process it
 - C. The client must maintain a persistent TCP connection with the server
 - D. Responses are cached indefinitely on the client side
62. Your service needs to notify an external system whenever a new user registers. Which mechanism is most suitable?
- A. You send a webhook POST to the external system's URL when registration occurs
 - B. The external system polls your API every minute
 - C. The external system reads a shared database table
 - D. You embed the external system's API key in the registration page
63. Which statement about SOAP is correct?
- A. SOAP messages are XML documents that contain an Envelope element
 - B. SOAP is a JSON-based protocol
 - C. SOAP does not support any form of security
 - D. SOAP messages are always sent using the HTTP GET method

64. Your API client must include credentials in each request to a protected REST endpoint. Which HTTP header is typically used for this purpose?
- A. Content-Type
 - B. Authorization
 - C. Cache-Control
 - D. User-Agent
65. You need to update only the email address field of an existing user resource. Which HTTP method should you use?
- A. PUT
 - B. PATCH
 - C. POST
 - D. DELETE
66. You are using Postman to create a new user via a POST request. The request body is JSON. Which header must you include to correctly describe the body format?
- A. Content-Type: application/json
 - B. Content-Type: multipart/form-data
 - C. Content-Type: text/plain
 - D. Content-Type: application/xml
67. After sending a GET request for a device, the API returns a 404 status code. What is the most probable reason?
- A. The requested resource URL does not exist on the server.
 - B. The server is overloaded and cannot process the request.
 - C. The request payload is malformed.
 - D. The client is not authorized to view the resource.
68. Your client receives a 202 HTTP status after submitting a data processing job. What does this status indicate?
- A. The request has been accepted for processing but is not yet complete.
 - B. The request was malformed and cannot be processed.
 - C. The server encountered an internal error while handling the request.
 - D. The request was successful and the resource was created.

69. An API supports pagination via a query parameter named "page". To retrieve the third page of results, which request URL is correct?
- A. GET /api/devices?page=3
 - B. GET /api/devices/page=3
 - C. GET /api/devices?offset=3
 - D. GET /api/devices?limit=3
70. You want the server to compress the response using gzip. Which request header should you add?
- A. Content-Encoding: gzip
 - B. Transfer-Encoding: gzip
 - C. Accept-Encoding: gzip
 - D. Cache-Control: gzip
71. A service uses the token bucket algorithm with a bucket capacity of 10 tokens and refills 1 token per second. If the client currently has 5 tokens and makes 3 requests in quick succession, how many tokens remain?
- A. 2 tokens
 - B. 8 tokens
 - C. 5 tokens
 - D. 10 tokens
72. Which authentication mechanism transmits the credentials as a Base64-encoded "username:password" string?
- A. API Key authentication
 - B. OAuth 2.0
 - C. Bearer authentication
 - D. Basic authentication
73. Your application must receive Cisco DNA Center webhook notifications about device alerts. Which of the following is NOT a required step to successfully consume the webhook?
- A. Handle incoming POST notifications appropriately
 - B. Ensure the application is continuously running to accept HTTP POST requests
 - C. Implement OAuth2 token refresh for the webhook endpoint
 - D. Register a URI with the webhook provider

74. After adding a new webhook subscription, your API call returns HTTP 401. What does this status code indicate?
- A. The server could not understand the request method
 - B. The client is not authorized to access the resource
 - C. The request payload is malformed
 - D. The requested resource was not found
75. Your webhook endpoint receives HTTP 415 when sending a JSON payload. Which of the following is the most probable reason?
- A. The request URL contains a typo
 - B. The client exceeded the rate limit
 - C. The server requires basic authentication
 - D. The Content-Type header is set to "application/xml" instead of "application/json"
76. When a script makes a request to a webhook URL and you see a long silence followed by a traceback, which side of the problem does this most likely indicate?
- A. Server-side connectivity or availability issue
 - B. Incorrect HTTP method used
 - C. Network firewall blocking the request
 - D. Client-side configuration error
77. You need to test how your system reacts when two users try to update the same resource via a webhook. Which HTTP status code should you simulate to represent this conflict?
- A. 200 OK
 - B. 403 Forbidden
 - C. 500 Internal Server Error
 - D. 409 Conflict
78. Your application sends rapid acknowledgments to a webhook provider and receives HTTP 429 responses. What does this status code mean?
- A. Bad Request – malformed payload
 - B. Gateway Timeout – upstream server did not respond
 - C. Forbidden – insufficient permissions
 - D. Too Many Requests – rate limit exceeded

79. A webhook POST is routed through a reverse proxy and the client receives HTTP 502. Which scenario most likely caused this response?
- A. The client failed to provide valid authentication credentials
 - B. The request payload was larger than the proxy's limit
 - C. The webhook URL was incorrectly spelled
 - D. The upstream service behind the proxy returned an invalid response
80. You run a Python script that calls the MapQuest Directions API and receive an HTTP 401 Unauthorized response. What is the most likely cause?
- A. Your API key is invalid or missing
 - B. Your network connection is down
 - C. The request payload is too large
 - D. The API endpoint URL is misspelled
81. After moving the application to production, API calls fail with SSL verification errors. What quick fix can you apply in Python's requests library while planning a secure solution?
- A. Set verify=False in the request call
 - B. Increase the request timeout value
 - C. Switch the request URL from https to http
 - D. Disable the firewall on the server

Matching (Dropdown Style)

82. Match the RESTful API method with its CRUD function.

- A. ERASE
 - B. CREATE
 - C. UPDATE
 - D. READ
 - E. DELETE
- GET: _____
- POST: _____
- DELETE: _____
- PUT/PATCH: _____

83. Match the rate limit algorithm with its description.

- A. Each user gets a defined number of tokens per time increment
- B. Requests are compressed
- C. Counts requests from the beginning of the window to the current time
- D. Requests enter a queue and are processed at a fixed rate
- E. A counter is assigned to a fixed window of time

Leaky bucket: _____

Token bucket: _____

Fixed window counter: _____

Sliding window counter: _____

84. Match the HTTP status code with its meaning.

- A. Internal server error
- B. Service unavailable
- C. Understood but rejected by the server
- D. Resource path not found on the server
- E. Resource created
- F. No valid authentication credentials

401: _____

403: _____

404: _____

500: _____

503: _____

85. Match the REST constraint with its description.

- A. Each layer serves only the layer below
- B. Responses must state whether they are cacheable
- C. Optional; responses may include executable code
- D. The server cannot contain session states
- E. Client and server are independent of each other

Stateless: _____

Cache: _____

Client-server: _____

Code-on-demand: _____

86. Match the SOAP element with its role.

- A. The prologue
- B. Error and/or status information
- C. The root element of the XML document
- D. Application-specific information such as authorization
- E. The data to be transported to the recipient

Envelope: _____

Header: _____

Body: _____

Fault: _____

Answer Key

No.	Answer	Why
1	C. One piece of software to talk to another	What is an API?
2	C. Automation tasks	Why use APIs?
3	C. Directly, providing data immediately	Synchronous APIs
4	C. A response with no data	Asynchronous APIs
5	A. The application can continue without being blocked while the server processes the request	Asynchronous APIs
6	D. RPC, SOAP, and REST	API Architectural Styles
7	A. Remote Procedure Call (RPC)	RPC slide
8	C. Envelope, Header, Body, and Fault	SOAP slide
9	D. Neutral	SOAP slide
10	B. Roy Thomas Fielding	REST slide
11	B. Six	REST slide
12	B. Stateless	REST: stateless
13	C. Cache	REST: cache
14	D. Hypermedia as the engine of application state	REST: uniform interface
15	A. Code-on-demand	REST: code-on-demand
16	B. URI, HTTP method, header, and body	REST API Requests
17	A. Scheme	URI slide
18	B. Query	URI slide
19	A. Path	URI slide
20	C. POST	HTTP method table
21	B. PATCH	HTTP method table
22	B. POST, PUT, and PATCH	Body slide
23	B. Content-Type	Body slide
24	C. HTTP status, header, and body	REST API Responses
25	C. Informational	HTTP status categories
26	D. Redirection	HTTP status categories
27	C. 201 Created	Common status codes
28	B. 403 Forbidden	Common status codes
29	C. 503 Service Unavailable	Common status codes
30	B. Set-Cookie	Response headers
31	D. Cache-Control	Response headers
32	C. A query parameter	Response Pagination
33	C. Accept-Encoding	Compressed response data
34	B. The user's identity	Authentication vs Authorization
35	A. The user's access to perform the action on a resource	Authentication vs Authorization

36	B. Bearer authentication	Authentication mechanisms
37	C. Public and private	Authentication mechanisms
38	D. It is not backwards compatible with OAuth 1.0	Authorization Mechanisms
39	B. A way for a web service to control how many requests a user or application can make per unit of time	What are Rate Limits?
40	C. Protect against a Denial-of-Service (DoS) attack	What are Rate Limits?
41	A. Leaky bucket	Leaky bucket
42	C. Token bucket	Token bucket
43	B. Fixed window counter	Fixed window counter
44	A. Sliding window counter	Sliding window counter
45	C. X-Rate Limit-Remaining	Knowing the Rate Limit
46	D. 429 Too Many Requests or 403 Forbidden	Exceeding the Rate Limit
47	B. Applications subscribe to the webhook server by registering with the provider	What is a Webhook?
48	B. It must register a URI on the webhook provider	Consuming a Webhook
49	C. MissingSchema	Troubleshooting: invalid URI
50	B. verify=False	Troubleshooting: invalid certificate
51	C. 407 Proxy Authentication Required	2xx and 4xx status codes
52	C. 409	2xx and 4xx status codes
53	B. 415 Unsupported Media Type	2xx and 4xx status codes
54	C. 504 Gateway Timeout	5xx status codes
55	D. 501 Not Implemented	5xx status codes
56	A. Check the return code, check the response body, use a status code reference	Interpreting Status Codes
57	D. No, contact the administrator of the API server	No response from API server
58	A. 4xx client-side, 5xx server-side	Summary
59	D. GET	Scenario question; matches the slide wording
60	C. Asynchronous API	Scenario question; matches the slide wording
61	B. Each request contains all information needed to understand and process it	Scenario question; matches the slide wording
62	A. You send a webhook POST to the external system's URL when registration occurs	Scenario question; matches the slide wording
63	A. SOAP messages are XML	Scenario question; matches the slide wording

	documents that contain an Envelope element	
64	B. Authorization	Scenario question; matches the slide wording
65	B. PATCH	Scenario question; matches the slide wording
66	A. Content-Type: application/json	Scenario question; matches the slide wording
67	A. The requested resource URL does not exist on the server.	Scenario question; matches the slide wording
68	A. The request has been accepted for processing but is not yet complete.	Scenario question; matches the slide wording
69	A. GET /api/devices?page=3	Scenario question; matches the slide wording
70	C. Accept-Encoding: gzip	Scenario question; matches the slide wording
71	A. 2 tokens	Scenario question; matches the slide wording
72	D. Basic authentication	Scenario question; matches the slide wording
73	C. Implement OAuth2 token refresh for the webhook endpoint	Scenario question; matches the slide wording
74	B. The client is not authorized to access the resource	Scenario question; matches the slide wording
75	D. The Content-Type header is set to "application/xml" instead of "application/json"	Scenario question; matches the slide wording
76	A. Server-side connectivity or availability issue	Scenario question; matches the slide wording
77	D. 409 Conflict	Scenario question; matches the slide wording
78	D. Too Many Requests – rate limit exceeded	Scenario question; matches the slide wording
79	D. The upstream service behind the proxy returned an invalid response	Scenario question; matches the slide wording
80	A. Your API key is invalid or missing	Scenario question; matches the slide wording
81	A. Set verify=False in the request call	Scenario question; matches the slide wording
82	GET = D; POST = B; DELETE = E; PUT/PATCH = C	HTTP method table
83	Leaky bucket = D; Token bucket = A; Fixed window counter = E; Sliding window counter = C	Rate Limit Algorithms slides
84	401 = F; 403 = C; 404 = D; 500 = A; 503 = B	Common status codes
85	Stateless = D; Cache = B; Client-server = E; Code-on-demand = C	REST constraints slides
86	Envelope = C; Header = D; Body = E; Fault = B	SOAP slide